Ekaterina Antonenko

Employment

since 2023 **Postdoctoral researcher**, *Centre for Computational Biology (CBIO) of École des Mines and Institut Curie (Paris, France)*.

2020–2023 **Data scientist**, *Digitalent (Paris, France)*.
CIFRE PhD project with École Polytechnique

2017–2019 **Math instructor**, *Russian School of Mathematics (New York, USA)*.

2013–2016 **Analyst**, *Moscow Center for Continuous Mathematical Education (Moscow, Russia)*.

Education

2020–2023 **École Polytechnique**, *PhD*.
Data Science and Mining (DaSciM) team, Laboratoire d'informatique (LIX), CIFRE with Digitalent
Scientific advisor: Jesse Read
Multi-target learning and prediction: novel methods and applications

2012–2014 **Moscow Bioinformatics School**.

2009–2014 **Moscow State University**, *Diploma in Mathematics*.

Papers in preparation

1. E. Antonenko, L. De Oliveira, A. Petit, M. Dogo, J. Cohen, S. Caillaud, P. Baduel, C. Azencott, V. Colot, *The DNA methylation lifespan of Transposable Elements in Arabidopsis Thaliana*.
2. E. Antonenko, G. Aguirre-Samboni, F. Massip, C. Azencott, *Integrative gene network analysis of genome-wide association data in Myalgic Encephalomyelitis / Chronic Fatigue Syndrome*.

Publications

3. E. Antonenko, A.Carreño, J. Read, *Autoreplicative Random Forests for missing value imputation*, Machine Learning journal (ECML/PKDD 2024 Journal Track), Volume 113, pages 7617-7643, 2024, https://link.springer.com/article/10.1007/s10994-024-06584-1.
4. E. Antonenko, M. Mechenich, R. Beigaitė, I. Žliobaitė, J. Read, *Backward inference in probabilistic Regressor Chains with distributional constraints*, Advances in Intelligent Data Analysis XXII - 22nd International Symposium, IDA 2024, https://link.springer.com/chapter/10.1007/978-3-031-58553-1_4.
5. E. Antonenko, J. Read, *Chains of Autoreplicative Random Forests for missing value imputation in high-dimensional datasets*, [Best paper award], Multi-Label Learning workshop at the ECML conference, 2022, https://mll2022.csd.auth.gr/papers/ChARF%20camera-ready.pdf.
6. E. Antonenko, J. Read, *Multi-modal ensembles of regressor chains for multi-output prediction*, Advances in Intelligent Data Analysis XX - 20th International Symposium, IDA 2022, https://link.springer.com/chapter/10.1007/978-3-031-01333-1_1.

7. V. Ivanenko, E. Antonenko, M. Gelfand, J. Yager, F. Ferrari, *Changes in segmentation and setation along the anterior/posterior axis of the homonomous trunk limbs of a remipede (Crustacea, Arthropoda)*, PeerJ, 2016, https://peerj.com/articles/2305/.
8. E. Antonenko, E. Yarovaya, *On the number of positive eigenvalues of the evolutionary operator of branching random walk*, Branching Processes and Their Applications, Lecture Notes in Statistics, vol. 219, Springer, Cham, pp. 41-55, 2016, https://doi.org/10.1007/978-3-319-31641-3_3.

Oral contributions

1. [Invited talk] *Beyond classical GWAS: epigenetic and network approaches to the genetics of complex traits*, Bioinfo seminar, Laboratoire Interdisciplinaire des Sciences du Numérique (LISN), Gif-sur-Yvette, France, October 2026.
2. [Poster] *Integrative gene network analysis of genome-wide association data in Myalgic Encephalomyelitis / Chronic Fatigue Syndrome*, European Conference on Computational Biology (ECCB 2026), Geneva, Switzerland, September 2026.
3. [Poster] *Integrative gene network analysis of genome-wide association data in Myalgic Encephalomyelitis / Chronic Fatigue Syndrome*, JOBIM 2026, Strasbourg, France, July 2026.
4. [Poster] *Understanding methylation of transposable elements in A. Thaliana via interpretable machine learning*, SMPGD 2026: Statistical Methods for Post Genomic Data, Grenoble, France, January 2026.
5. [Talk] *From biology to machine learning and back: understanding transposable element methylation and its phenotypic effects*, Paris Postdocs Seminar, Institute Imagine, Paris, France, January 2026.
6. [Poster] *Spreading of methylation of transposable elements in A. Thaliana*, AI workshop at École des Mines, Paris, France, December 2025.
7. [Talk] *Spreading of methylation of transposable elements in A. Thaliana*, Meeting LEGO 2025 – Machine learning for genomics, days of GDR BIMMM, Nantes, France, November 2025.
8. [Poster] *Genotype Imputation with Multi-label Random Forests*, ISMB/ECCB 2023, MLCSB Cosi, Lyon, France, July 2023.
9. [Talk] *Multi-output machine learning with applications to genomics*, Institut Imagine, Paris, France, May 2023.
10. [Invited talk] *Autoreplicative Random Forests for missing value imputation*, Group seminar, KU Leuven KULAK, Kortrijk, Belgium, February 2023.
11. [Poster] *Genotype imputation with multi-label Random Forests*, Machine Learning in Computational Biology, online, November 2022.
12. [Talk] *Chains of Autoreplicative Random Forests for missing value imputation in high-dimensional datasets*, Multi-Label Learning workshop: current trends and open challenges, ECML PKDD 2022, Grenoble, France, September 2022.
13. [Talk] *Multi-modal ensembles of Regressor Chains for multi-output prediction*, Intelligent Data Analysis XXI - 21st International Symposium, IDA 2022, Rennes, France, April 2022.

Teaching

- Machine learning and statistics for genomics, Seminars, *PSL (M2)*,
  - Fall 2025
- Machine Learning, Seminars and mini-projects, *École des Mines (M1/M2)*,
  - Fall 2026, 2025, 2023
- Data science (Science des données), Seminars (in French), *École des Mines (L3)*,
  - Summer 2026, 2025
- LSML: Large-Scale Machine Learning & Data Mining, Lecture on Introduction to large-scale ML & optimization, *École des Mines (M1/M2)*,
  - March 2026, 2025, 2024
- CSE204 Machine Learning, Seminars, *École Polytechnique (L2)*,
  - Spring 2023, 2022

Mentoring students

- Alix Gillet (Université de Rouen Normandie), M1 internship *"Studies of the pan-genomic association of transposable elements polymorphisms and epigenetic regulation in Arabidopsis thaliana"*, co-advised with Chloé-Agathe Azencott, 2026.
- Marie Dogo (École des Mines), M1 internship *"Spreading of methylation in Transposable Elements"*, co-advised with Chloé-Agathe Azencott, 2025.
- Jérémy Cohen (École Centrale de Nantes), M2 internship *"Interpretable models for the methylation of Transposable Elements"*, co-advised with Chloé-Agathe Azencott, 2024.
- Margarita Konnova (École Polytechnique), Bachelor thesis *"Missing Value Imputation for Genomics Data using a Sequence Based Generative Adversarial Network"*, co-advised with Jesse Read, 2022.

Thesis defense committees

- Frédérick Fabre Ferber, *Petits jeux de données et prédiction en Intelligence Artificielle, vers une meilleure cohabitation : Application à la gestion durable de l'enherbement des systèmes agricoles à La Réunion*, l'Université de La Réunion (France), 2025.

Reviewing

- Conference reviewer: ECML PKDD Research Track (2026, 2025, 2024), ISMB - MLCSB track (2026), MLCB (2026, 2025, 2024, 2023).
- Journal reviewer: Machine Learning, Data Mining and Knowledge Discovery, Communications Biology.

Additional training

- Data Science: Multiple Imputation in Practice, *Utrecht University, The Netherlands*, 2022.
- AI for Medicine Specialization, *Deeplearning.ai*, 2021, *Certificate*.
- Deep Learning Specialization, *Deeplearning.ai*, 2020, *Certificate*.

- Machine Learning, *Stanford University*, 2019, *Certificate*.
- Molecular Biology – Part 3: RNA Processing and Translation, *Massachusetts Institute of Technology*, 2019, *Certificate*.
- Molecular Biology – Part 2: Transcription and Transposition, *Massachusetts Institute of Technology*, 2019, *Certificate*.
- Molecular Biology – Part 1: DNA Replication and Repair, *Massachusetts Institute of Technology*, 2018, *Certificate*.
- Quantitative Biology Workshop, *Massachusetts Institute of Technology*, 2018, *Certificate*.
- Introduction to Biology – The Secret of Life, *Massachusetts Institute of Technology*, 2018, *Certificate*.
- Bioinformatics Algorithms (Part 1), *University of California, San Diego*, 2014, *Certificate*.

Additional skills

- Programming: Python, R, bash
- Languages: English, French, Russian